

Summary –D2

Conditional Statements

Let p and q be propositions.

- The *conditional statement* $p \Rightarrow q$ is the proposition “if p , then q .”

Core Components

- The hypothesis: The initial assumption (p).
 - The conclusion: The result that follows (q).
 - The implication: The logical connection between them ($p \Rightarrow q$).
- The conditional statement $p \Rightarrow q$ is false when p is true and q is false, and true otherwise.
- In the conditional statement $p \Rightarrow q$, p is called the ***hypothesis*** (or *antecedent* or *premise*) and q is called the ***conclusion*** (or *consequence*).
- The fundamental logic of mathematical implication ($\Rightarrow$), which is structured as "If [**hypothesis**], then [**conclusion**]."

Common ways to express this conditional statement are:

- “if p , then q ”
- “ p implies q ”
- “if p , q ”
- “ p only if q ”
- “ p is sufficient for q ”
- “a sufficient condition for q is p ”
- “ q if p ”
- “ q whenever p ”
- “ q when p ”
- “ q is necessary for p ”
- “a necessary condition for p is q ”
- “ q follows from p ”
- “ q unless $\neg p$ ”

Converse, Contrapositive, and Inverse

We can form some new conditional statements starting with a conditional statement

$p \Rightarrow q$. In particular, there are three related

conditional statements that occur so often that they have special names.

- The proposition $q \Rightarrow p$ is called the **converse** of $p \Rightarrow q$.
- The proposition $\neg q \Rightarrow \neg p$ is called the **contrapositive** of $p \Rightarrow q$.
- The proposition $\neg p \Rightarrow \neg q$ is called the **inverse** of $p \Rightarrow q$.

We will see that of these three conditional statements formed from $p \Rightarrow q$, only the contrapositive always has the same truth value as $p \Rightarrow q$.

We first show that the contrapositive, $\neg q \Rightarrow \neg p$, of a conditional statement $p \Rightarrow q$ always has the same truth value as $p \Rightarrow q$.

To see this, note that the contrapositive is false only when $\neg p$ is false and $\neg q$ is true, that is, only when p is true and q is false.

We now show that neither the converse, $q \Rightarrow p$, nor the inverse, $\neg p \Rightarrow \neg q$, has the same truth value as $p \Rightarrow q$ for all possible truth values of p and q .

Note that when p is true and q is false, the original conditional statement is false, but the converse and the inverse are both true.

Logically Equivalent

Definition: When two compound propositions always have the same truth value we call them **equivalent**.

The symbol $\equiv$ is used to denote the logical equivalences.

For example, $p \equiv q$ denotes the two compound propositions p and q are logically equivalent.

Note that the symbol $\equiv$ is not a logical connective, and $p \equiv q$ is not a compound proposition.

Note that a conditional statement and its contrapositive are equivalent. The converse and the inverse of a conditional statement are also equivalent, as the reader can verify, but neither is equivalent to the original conditional statement.

The contrapositive and converse appear quite frequently in mathematical writing, but the inverse is rare (in this author's experience at least). The truth tables of the implication, contrapositive, converse and inverse are:

c_1	c_2	c_3	c_4	c_5	c_6	c_7	c_8
p	q	$\sim p$	$\sim q$	$p \Rightarrow q$	$\sim q \Rightarrow \sim p$	$q \Rightarrow p$	$\sim p \Rightarrow \sim q$
T	T	F	F	T	T	T	T
F	T	T	F	T	T	F	F
T	F	F	T	F	F	T	T
F	F	T	T	T	T	T	T

By c_5 and c_6 we set $p \Rightarrow q \equiv \sim q \Rightarrow \sim p$ By c_7 and c_8 we have $q \Rightarrow p \equiv \sim p \Rightarrow \sim q$

The above tables show that the original implication and the contrapositive have the exactly same truth tables, and that the converse and inverse have the same tables. However we also see that the original implication does not have the same table as the converse or inverse. The inverse is not very commonly used, however the contrapositive and converse will be very useful for us as we continue.

Example: "If x is a real number, then x^2 is a real number"

Write down:

(i) **Converse** of the above compound statement

if x^2 is a real number, then x is a real number

(ii) **Inverse** of the above compound statement

if x is not a real number, then x^2 is not a real number

(iii) **Contrapositive** of the above compound statement

if x^2 is not a real number, then x is not a real number

(iv) **Negation** of the above the above compound statement

$\sim(p \Rightarrow q) ??$ $\sim(p \Rightarrow q) \equiv p \wedge \sim q$

" x is a real number and x^2 is not a real number."

Biconditional

Definition: The biconditional. Let p and q be statements. The biconditional, $p \Leftrightarrow q$, read as “ p if and only if q ”, is true when p and q have the same truth value and false when p and q take different truth values.

The biconditional $p \Leftrightarrow q$ is also sometimes written as

- p iff q ,
- p is equivalent to q ,
- p is a necessary and sufficient condition for q
- If p , then q , and conversely.

Note that “iff” is still read as “if and only if” (and not as “ifffftfffff” with a long “f”-noise).

Example: n^2 is odd if and only if n is odd.

The Truth Table for the Biconditional ($p \Leftrightarrow q$)

p	q	$p \Leftrightarrow q$
T	T	T
F	T	F
T	F	F
F	F	T

Note that $p \Leftrightarrow q$ has exactly the same truth value as $(p \Rightarrow q) \wedge (q \Rightarrow p)$.

p	q	$p \Leftrightarrow q$	$p \Rightarrow q$	$q \Rightarrow p$	$(p \Rightarrow q) \wedge (q \Rightarrow p)$
T	T	T	T	T	T
F	T	F	T	F	F
T	F	F	F	T	F
F	F	T	T	T	T

$$p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge (q \Rightarrow p)$$

Implicit use of biconditionals

You should be aware that biconditionals are not always explicit in natural language. In particular, the “if and only if” construction used in biconditionals is rarely used in common language. Instead, biconditionals are often expressed using an “if, then” or an “only if” construction. The other part of the “if and only if” is implicit. That is, the converse is implied, but not stated. For example, consider the statement in English

“If you finish your meal, then you can have dessert.”

What is really meant is *“You can have dessert if and only if you finish your meal.”* This last statement is logically equivalent to the two statements

“If you finish your meal, then you can have dessert” and *“You can have dessert only if you finish your meal.”*

Because of this imprecision in natural language, we need to make an assumption whether a conditional statement in natural language implicitly includes its converse. Because precision is essential in **mathematics** and in **logic**, we will always distinguish between the conditional statement $p \Rightarrow q$ and the biconditional statement $p \Leftrightarrow q$.

Exercise:

1. Let p, q , and r be statements. Suppose that

- “ $p \Rightarrow (q \wedge r)$ ” is false, and
- “ $(\sim q) \wedge r \Rightarrow (\sim p)$ ” is true.

Determine the truth values of p, q , and r .

Truth Tables of Compound Propositions

We have now introduced four important logical connectives, conjunctions, disjunctions, conditional statements, and biconditional statements as well as negations. We can use these connectives to build up complicated compound propositions involving any number of propositional variables. We can use truth tables to determine the truth values of these compound propositions. We use a separate column to find the truth value of each compound expression that occurs in the compound proposition as it is built up. The truth values of the compound proposition for each combination of truth values of the propositional variables in it is found in the final column of the table.

Examples:

1. Construct the truth table of the compound proposition $(p \vee \neg q) \Rightarrow (p \wedge q)$.

Because this truth table involves two propositional variables p and q , there are four rows in this truth table, one for each of the pairs of truth values TT, FT, TF, and FF. The first two columns are used for the truth values of p and q , respectively. In the third column we find the truth value of $\neg q$, needed to find the truth value of $p \vee \neg q$, found in the fourth column. The fifth column gives the truth value of $p \wedge q$. Finally, the truth value of $(p \vee \neg q) \Rightarrow (p \wedge q)$ is found in the last column. The resulting truth table is given below.

p	q	$\neg q$	$p \vee \neg q$	$p \wedge q$	$(p \vee \neg q) \Rightarrow (p \wedge q)$
T	T	F			
F	T	F			
T	F	T			
F	F	T			

2. Let p and q be statements. Write down the truth tables for

(a) $(\sim p) \Rightarrow q$

(b) $(p \wedge q) \vee (\sim p) \Rightarrow q$

(c) $p \wedge (\sim p)$

(d) $p \vee (\sim p)$

(e) $\sim(p \Leftrightarrow q)$

(a)

p	q	$\sim p$	$\sim p \Rightarrow q$
T	T	F	T
F	T	T	T
T	F	F	T
F	F	T	F

no. of rows

$$2^2 = 4 \text{ rows}$$

(c) no. of rows $2^1 = 2$

p	$\sim p$	$p \wedge \sim p$
T	F	F
F	T	F

Note:

To construct the truth table for a given proposition:

1. Create a table with 2^n rows if a compound proposition involves n propositions.
2. Fill in the first n columns with all possible combinations.
3. Determine and enter the truth value for each combination.

Example: For $n=3$

p	q	r
T	T	T
F	T	T
T	F	T
F	F	T
T	T	F
F	T	F
T	F	F
F	F	F

Precedence of Logical Operators

We can construct compound propositions using the negation operator and the logical operators defined so far. We will generally use parentheses to specify the order in which logical operators in a compound proposition are to be applied. For instance, $(p \vee q) \wedge (\neg r)$ is the conjunction of $p \vee q$ and $\neg r$. However, to reduce the number of parentheses, we specify that the negation operator is applied before all other logical operators. This means that $\neg p \wedge q$ is the conjunction of p and q , namely, $(\neg p) \wedge q$, not the negation of the conjunction of p and q , namely $\neg(p \wedge q)$.

Another general rule of precedence is that the conjunction operator takes precedence over the disjunction operator, so that $p \wedge q \vee r$ means $(p \wedge q) \vee r$ rather than $p \wedge (q \vee r)$. Because this rule may be difficult to remember, we will continue to use parentheses so that the order of the disjunction and conjunction operators is clear.

Finally, it is an accepted rule that the conditional and biconditional operators $\Rightarrow$ and $\Leftrightarrow$ have lower precedence than the conjunction and disjunction operators, $\wedge$ and $\vee$. Consequently, $p \vee q \Rightarrow r$ is the same as $(p \vee q) \Rightarrow r$. We will use parentheses when the order of the conditional operator and biconditional operator is at issue, although the conditional operator has precedence over the biconditional operator. Following table displays the precedence levels of the logical operators, $\neg$, $\wedge$, $\vee$, $\Rightarrow$, and $\Leftrightarrow$.

Precedence of Logical Operators.

Operator	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\Rightarrow$	4
$\Leftrightarrow$	5

Example: How do we describe the statement $\neg p \Rightarrow p \vee q \wedge \neg r$?

Tautologies and Contradictions

Definition:

A **tautology** is a compound statement S that is true for all possible combinations of truth values of the component statements that are part of S .

A **contradiction** is a compound statement S that is false for all possible combinations of truth values of the component statements that are part of S .

That is, a tautology is necessarily true in all circumstances, and a contradiction is necessarily false in all circumstances.

Examples: (Tautologies and Contradictions)

For statements p and q :

1. Use a truth table to show that $p \vee \neg p$ is a tautology.
2. Use a truth table to show that $p \wedge \neg p$ is a contradiction.
3. Use a truth table to determine if $p \Rightarrow p \vee q$ is a tautology, a contradiction, or neither

Note: Tautologies and Contradictions

- A compound proposition that is **always true**, no matter what the truth values of the propositions that occur in it, is called a **tautology**.
- A compound proposition that is **always false** is called a **contradiction**.
- A compound proposition that is neither a tautology nor a contradiction is called a **contingency**.

Recall: Logically equivalent

Two compound propositions are **logically equivalent** if they have **all the same truth values** for corresponding sets of truth values of the propositions that occur in them. The notation

$p \equiv q$ denotes that p and q are logically equivalent.

The following are some useful logical equivalences.

$$(i) \sim(\sim p) \equiv p$$

$$(ii) \sim(p \wedge q) \equiv \sim p \vee \sim q$$

$$(iii) \sim(p \vee q) \equiv \sim p \wedge \sim q$$

$$(iv) p \Rightarrow q \equiv \sim p \vee q$$

$$(v) \sim(p \Rightarrow q) \equiv p \wedge \sim q$$

End of D₃